Gamified Physiotherapy:

An Interactive Rehabilitation System Using AI and Motion Tracking for Brachial Plexus Injury

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Introduction

Brachial Plexus injuries affect upper limb mobility and require long, repetitive rehabilitation programs. To address limitations in traditional therapy approaches, an intelligent and adaptive rehabilitation system was developed to transforms physical exercises into interactive, game-based activities supported by Artificial Intelligence.

With a comprehensive medical dashboard that enables accurate monitoring of patient progress and provides clinicians with data-driven insights for decision-making, the system offers a complete, smart and effective rehabilitation solution.

Development Tools

JavaScript, CSS, HTML, C#, django, Python, MySQL, Unity, MediaPipe

Methodology

An iterative development methodology was followed to progressively design, implement, and validate the system through multiple development cycles. Each iteration included prototyping, testing, and refining system features.

Requirement Specification
Literature review & 16 physiotherapists consultations

Motion Tracking Integration
Development of rehabilitation games integrated with real-time body tracking

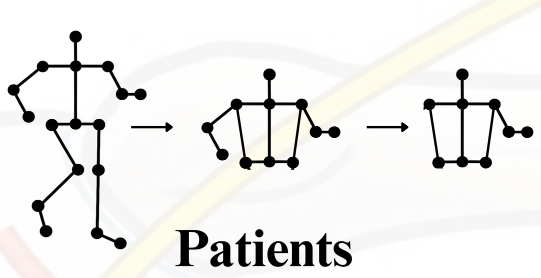
Intelligent Therapy Adaptation
Implementation of AI-based evaluation and adaptive rehabilitation logic

Clinical Monitoring Dashboard
Visualization of patient progress and data-driven clinical insights

Validation & Refinement
Iterative testing with patients and physiotherapists evaluation

System Design

Clinicians
Patient Registration & Profile Initialization (including clinical data and affected limb selection)



Patients
Voice-guided Initial Assessment using real-time video capture

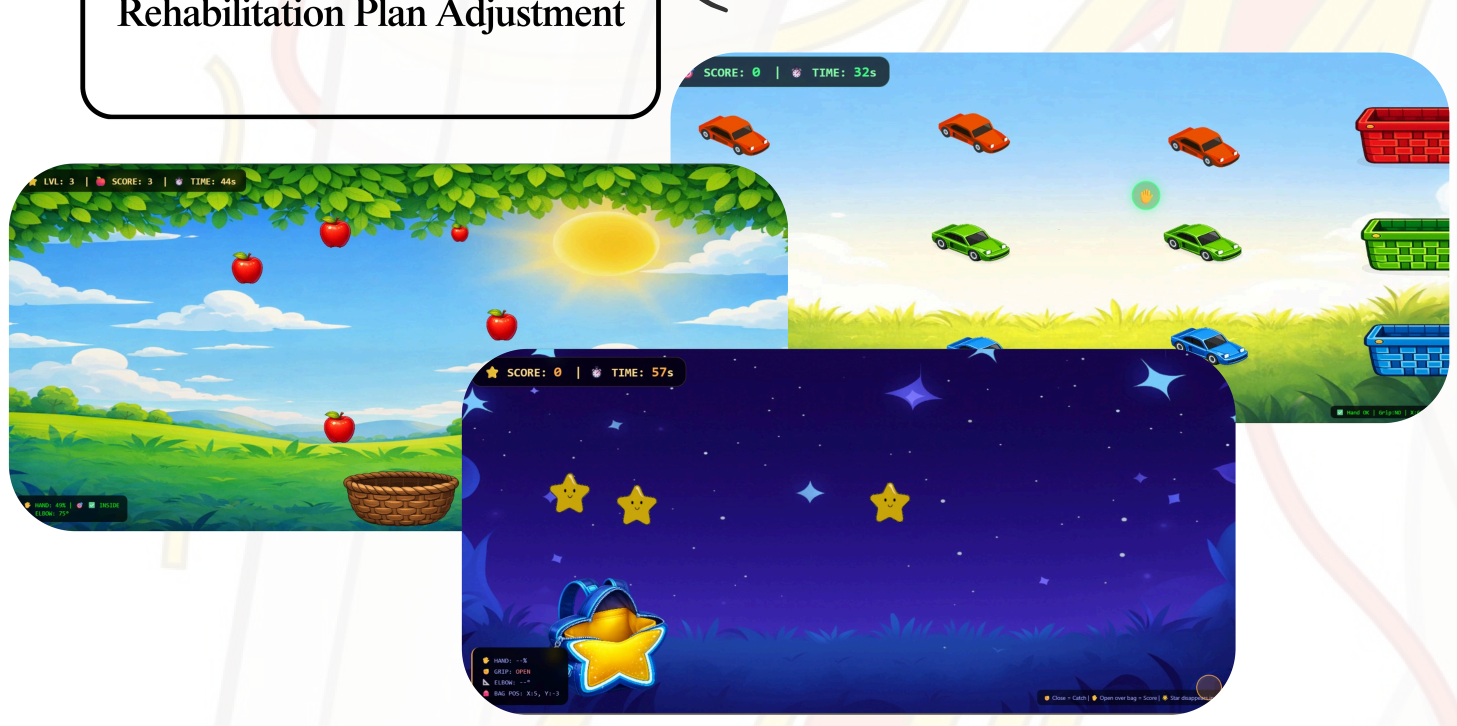
System
Body Landmark Detection & Joint Angle Computation using MediaPipe and geometric calculations

Patients
Adaptive Game-based Rehabilitation Exercises with motion mapped to in-game actions

System
Performance Evaluation and Score Calculation based on ROM accuracy and consistency

System
Session Data Storage & Progress Tracking with dashboard visualization and historical analysis

Clinicians
Progress Monitoring and Rehabilitation Plan Adjustment



Expirements and Results

The proposed system has been reviewed by physiotherapists, who indicated that it is expected to be effective and achieve promising rehabilitation outcomes. Formal evaluation was confirmed to be conducted through real-world long term deployment involving patients under the supervision of their physiotherapists across multiple physiotherapy clinics. Evaluation will not be limited to individuals with Brachial Plexus Injuries (BPI), but will also include patients with various conditions where rehabilitation exercises are deemed appropriate by clinicians, such as Stroke, Cerebral Palsy, and Post-surgical recovery cases. Participants will represent diverse genders and age groups, ranging from 4 to 52 years.

- ROM Improvement — 80% - 95%
- Task Completion Rate — 90% - 100%
- Patient Engagement — 97% - 100%
- System Adaptation Accuracy — 90% - 98%
- Clinician Approval — 100%

* Expected results under Physiotherapists approval

Physiotherapists positively evaluated the system and confirmed its practical usefulness in clinical settings.

Conclusion

This project demonstrates the effectiveness of integrating gamification and adaptive technologies into rehabilitation systems. The proposed solution improves patient engagement, supports personalized therapy, and encourages consistent participation in rehabilitation exercises.

By addressing key limitations of traditional rehabilitation methods, the system enhances the overall therapy experience and shows strong potential to improve recovery outcomes not only for patients with Brachial Plexus Injuries (BPI), but also for individuals with other rehabilitation-related conditions requiring long- or short-term physical therapy.

References:

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